

genaut — frontier report

source CSV : genaut/logs/1state3ap0acc/default.csv
corpus rows: 52

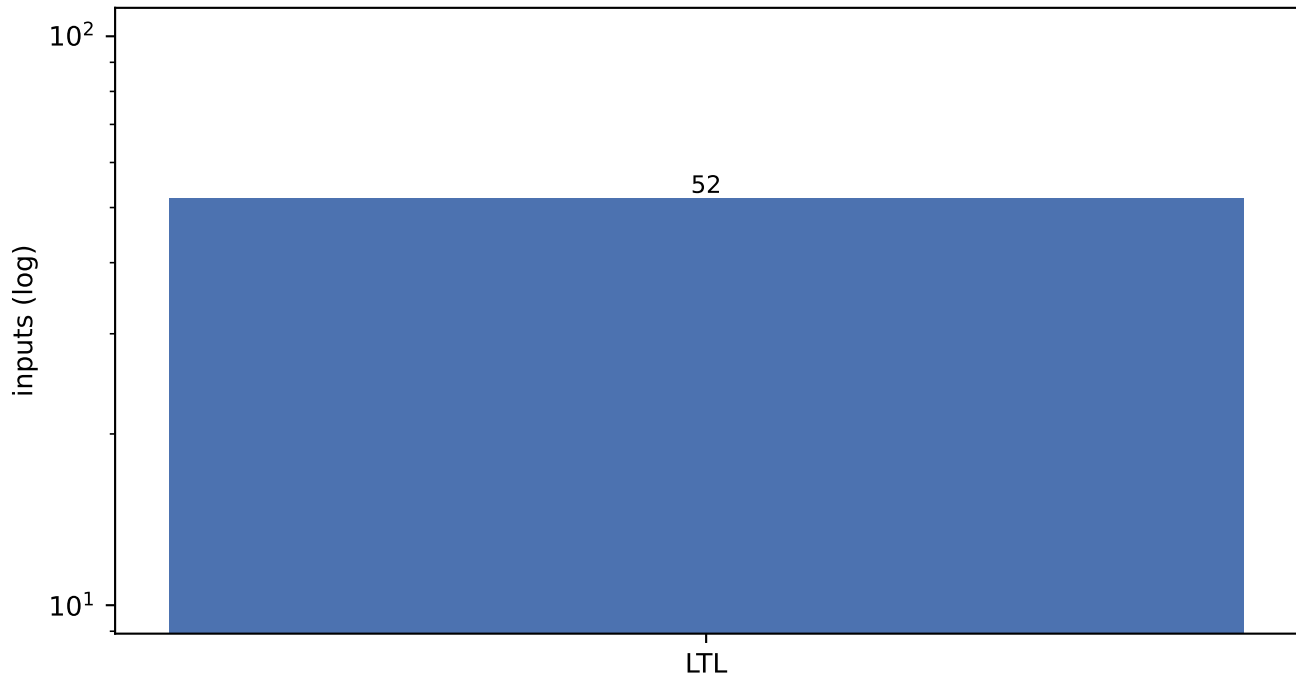
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 52
  LTL-definable (formula built) :    52  (100.0%)
  not-LTL (decided)             :      0  (  0.0%)
  ambiguous (timeout, unknown)  :      0  (  0.0%)
decided (LTL or not-LTL): 52  (100.0%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 52 (0 with a gated/unprintable string)

distinct structures (md5): 52

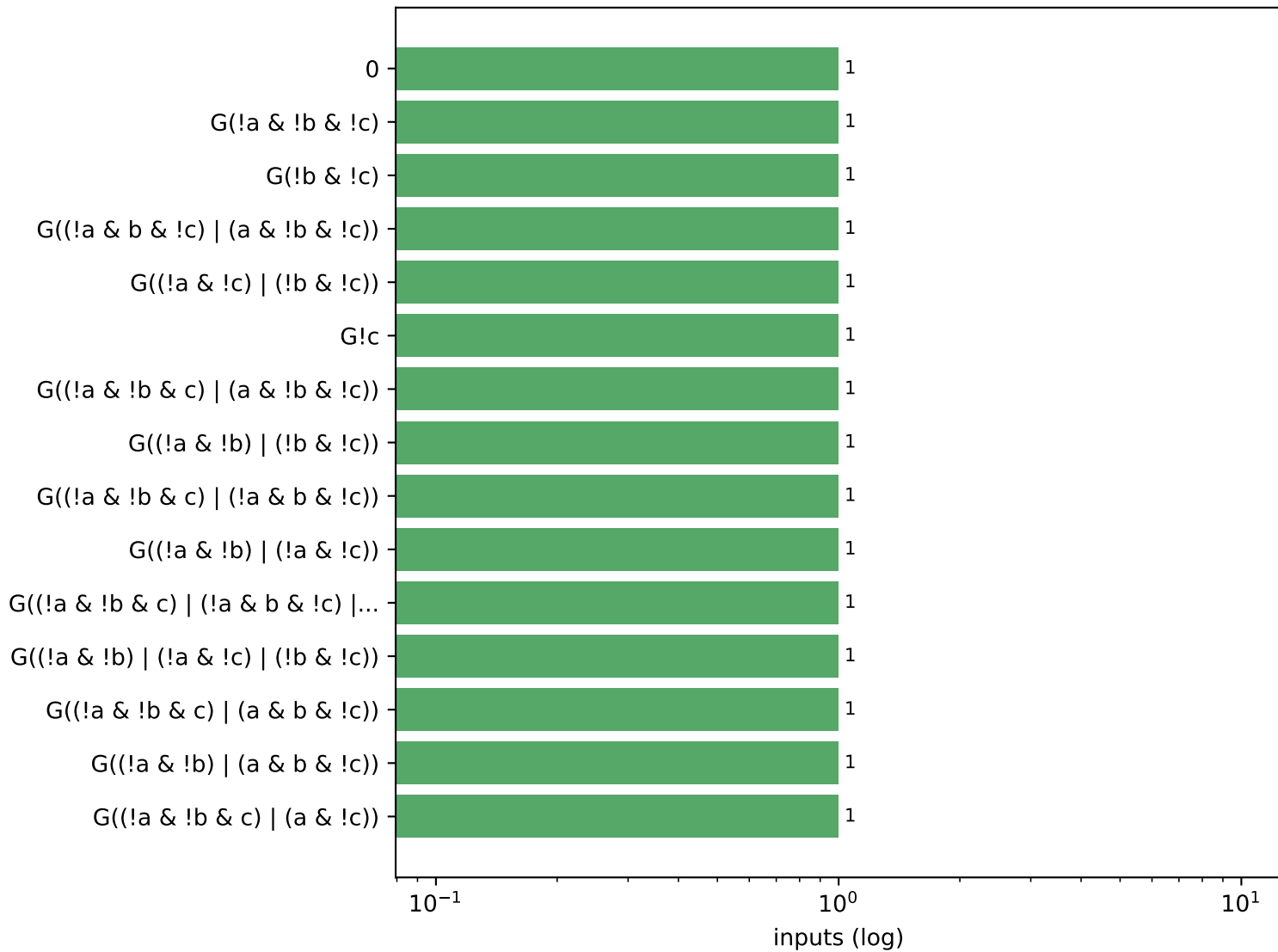
80% of answers covered by the top 42 structure(s)

90% of answers covered by the top 47 structure(s)

top 15 structures:

```
1 ( 1.9%) 0
1 ( 1.9%) G(!a & !b & !c)
1 ( 1.9%) G(!b & !c)
1 ( 1.9%) G(!a & b & !c) | (a & !b & !c)
1 ( 1.9%) G(!a & !c) | (!b & !c)
1 ( 1.9%) G!c
1 ( 1.9%) G(!a & !b & c) | (a & !b & !c)
1 ( 1.9%) G(!a & !b) | (!b & !c)
1 ( 1.9%) G(!a & !b & c) | (!a & b & !c)
1 ( 1.9%) G(!a & !b) | (!a & !c)
1 ( 1.9%) G(!a & !b & c) | (!a & b & !c) | (a & !b & !c)
1 ( 1.9%) G(!a & !b) | (!a & !c) | (!b & !c)
1 ( 1.9%) G(!a & !b & c) | (a & b & !c)
1 ( 1.9%) G(!a & !b) | (a & b & !c)
1 ( 1.9%) G(!a & !b & c) | (a & !c)
```

Top 15 reconstructed structures



Size frontier — flattened formula size

LTL answers with a size: 52

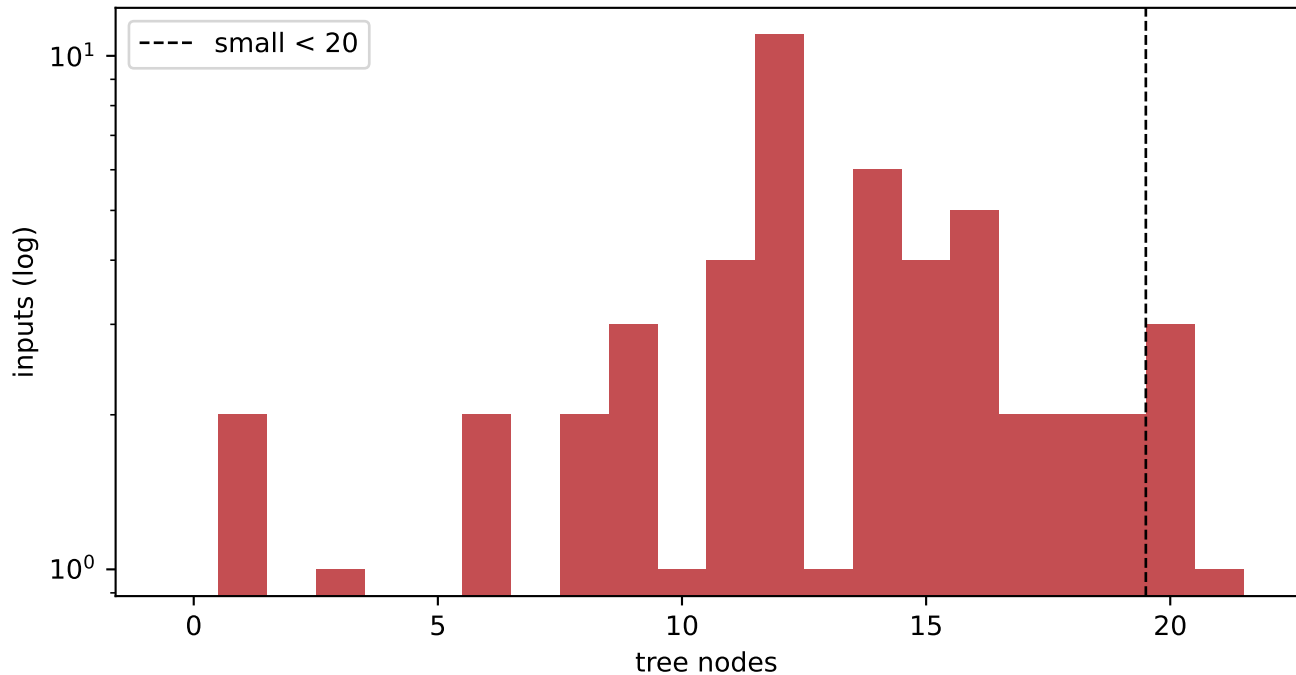
trivial (tree≤1, i.e. true/false): 2 (3.8%)

small (tree<20): 48 (92.3%)

larger (tree≥20): 4 (7.7%)

tree nodes: median 12 mean 13.0 max 24

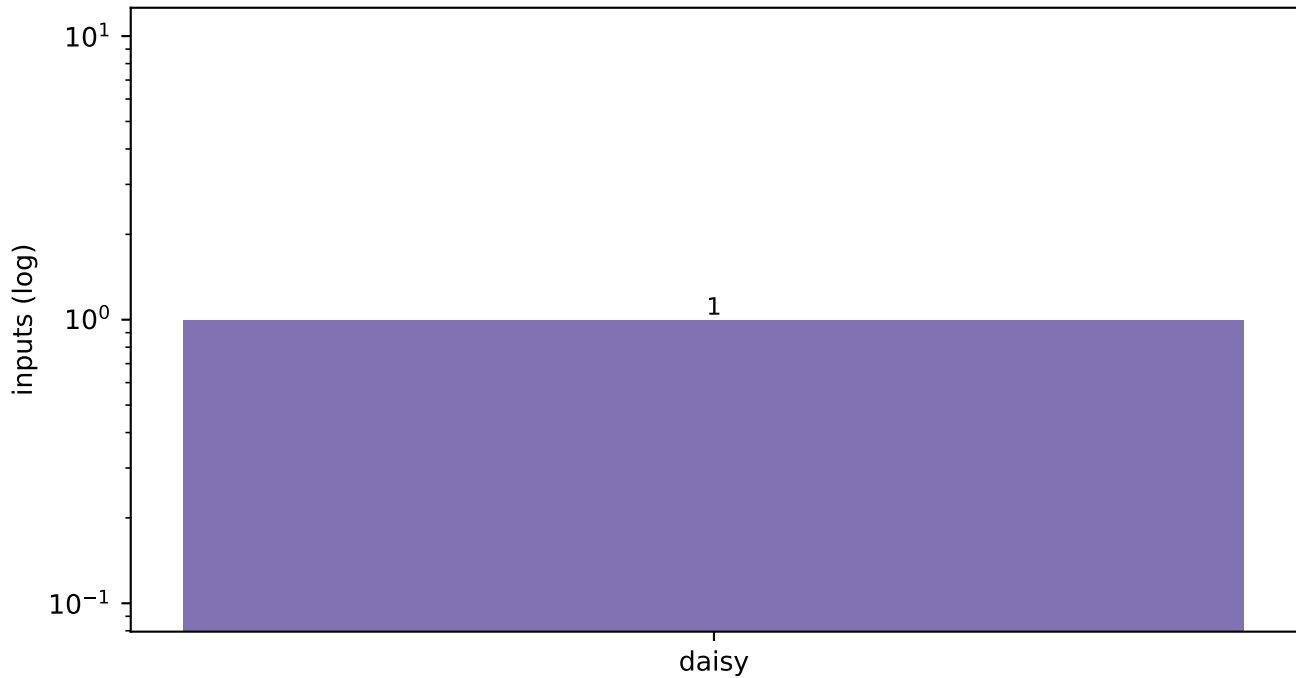
Flattened size (tree nodes); clipped at 21 (1 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 1
  1 (100.0%) daisy
build_s over true rows: total 0.1s  mean 0.068s  max 0.068s
```

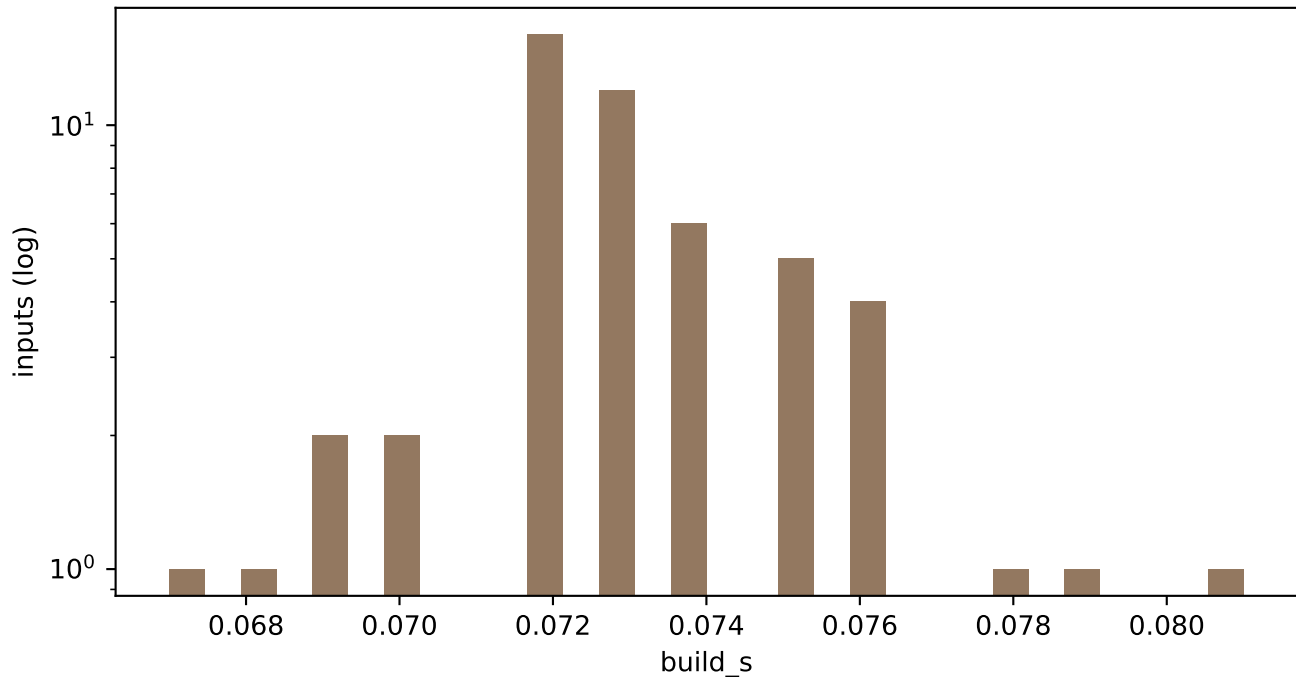

Routes to `true` (technique)



Build time & timeouts

```
total build: 3.8s   mean 0.073s   max 0.081s
  LTL          n= 52   mean 0.073s   max 0.081s
timeouts: 0
```

Build-time distribution



Verification — spot-oracle validation

validation over 52 LTL answers:
True 52 (100.0%)
NOT-equivalent (FALSE): 0 CLEAN